

WASP-43b

R-1099

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_160305 · created 2026-07-30T19:05:29+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457452.7336 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.058
Transit depth (Rp/Rs)^2	3.91 +/- 2.31 [%]
Orbital Inclination (inc)	86.93 +/- 2.84
Airmass coefficient 1 (a1)	7.72 +/- 4.35
Airmass coefficient 2 (a2)	-1.47 +/- 0.3
Scatter in the residuals of the lightcurve fit is	44.13 %
Best Comparison Star	None
Optimal Aperture	4.47
Optimal Annulus	22.36
Transit Duration (day)	0.0614 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.198 ± 0.058 vs 0.1594 (deviation 0.44σ)
Duration fit vs published	0.0614 d vs 0.0512 d (deviation 1.38σ)
Mid-time O–C	-1.9 min
Depth SNR	2.3
Residual scatter	44.13 %

Light curve

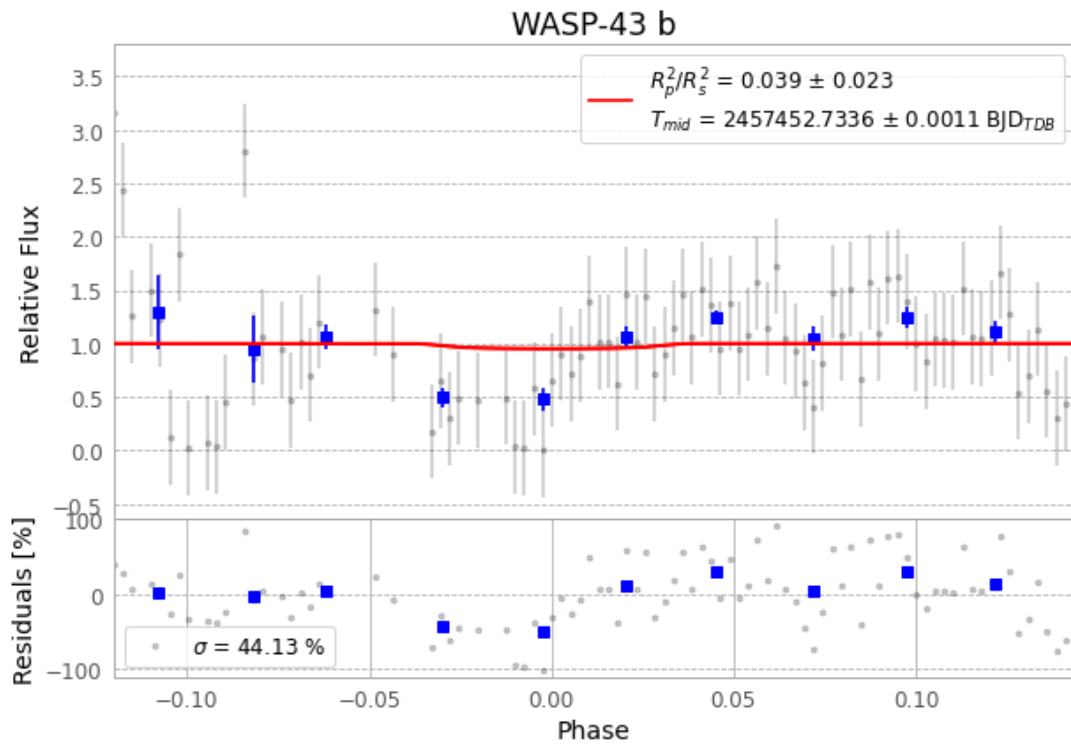


Figure 1 — FinalLightCurve_WASP-43 b_2016-03-04.png (SHA-256 6a679a6fb99c0c72...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [283, 139], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 069577933e1569e3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

104 FITS frames (68 MB), WASP-43160305030612.FITS → WASP-43160305081512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-160305.log (a207e1fef433...), FinalLightCurve_WASP-43 b_2016-03-04.png (6a679a6fb99c...), FinalLightCurve_WASP-43 b_2016-03-04.csv (8aa0dd47450d...)

Compute resources

Wall time 235.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 19:05Z.